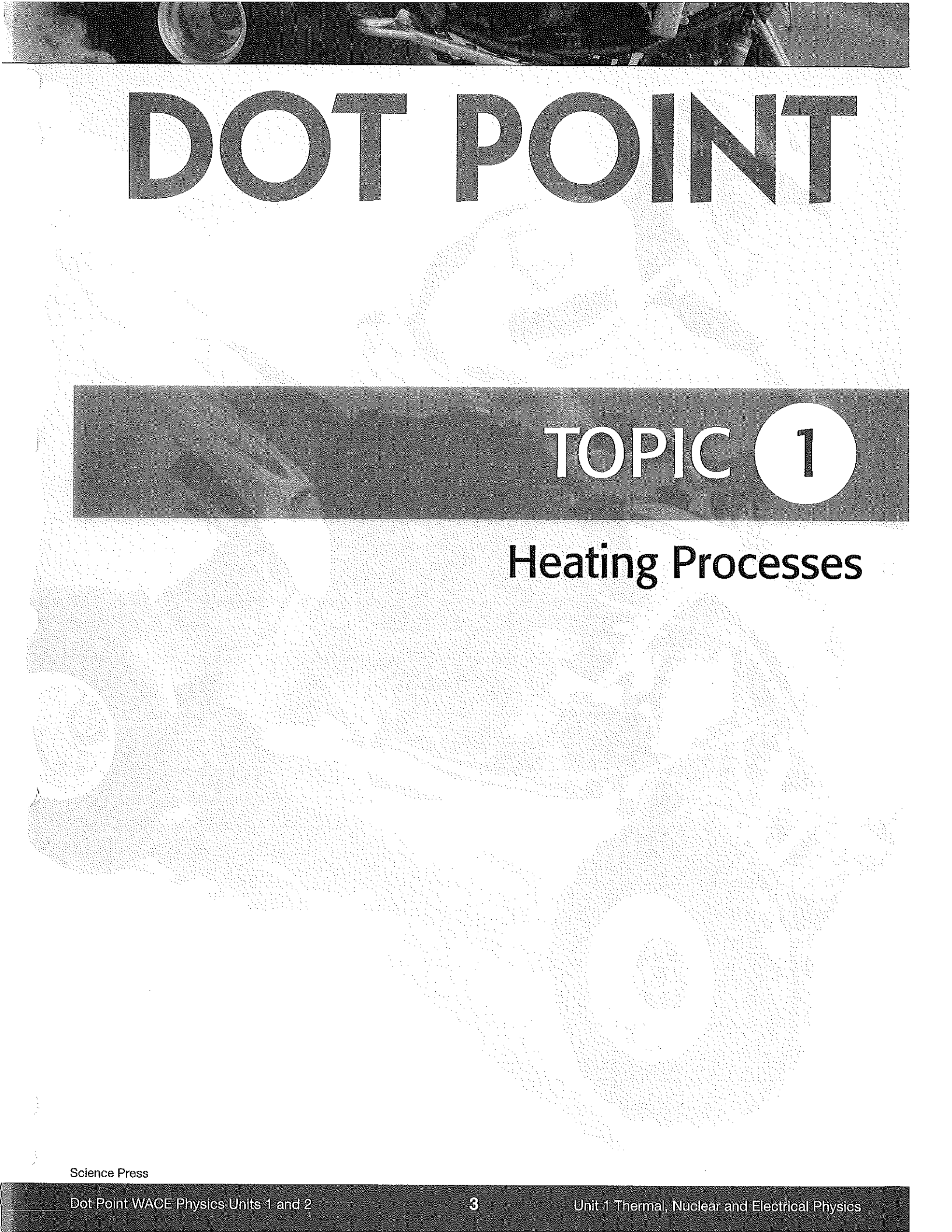




DOT POINT

TOPIC 1

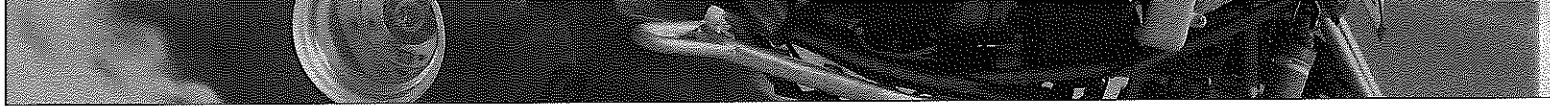
Heating Processes

1.1 The kinetic particle model, internal thermal energy and temperature.

1.1.1 The kinetic theory of matter.

1.1.1.1 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise the kinetic theory of matter.

Completed sentences	Second halves (not in order)
All matter is	at absolute zero.
The particles of matter are always moving except	they are held together by strong forces.
Particles in matter are held	in any direction so that gases always fill their containers.
The particles in solids cannot move freely because	are weaker than those holding particles together in solids.
Particles in solids simply vibrate	studying the properties of matter.
In liquids particles can	made up of particles which are in continuous motion.
The forces holding particles together in liquids	together by forces which vary in strength.
Particles in gases are free to move	roll over one another.
Particles in gases are	to explain and predict the behaviour of matter.
Particles of matter are far too	in their fixed positions.
We deduce the behaviour of the particles of matter by	small to be seen even under a microscope.
We use the kinetic theory of matter	not held together.

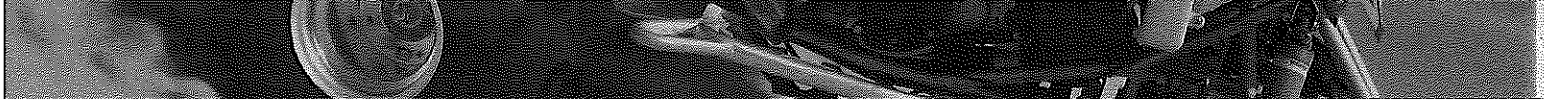


1.1.2 Kinetic theory and properties of matter.

- 1.1.2.1** Why can't we compress liquids and solids?
- (A) They are too dense.
 - (B) Their particles are held together too tightly.
 - (C) Their particles are further apart.
 - (D) Their particles are already as close together as they can be.
- 1.1.2.2** Why can liquids be poured?
- (A) Their particles are held together loosely.
 - (B) Their particles are not held together.
 - (C) Their particles can roll over one another.
 - (D) Their particles are held together tightly.
- 1.1.2.3** What will cause the particles of matter to move more slowly?
- (A) Weaker forces between them.
 - (B) Cooling the matter.
 - (C) Heating the matter.
 - (D) Shaking the matter.
- 1.1.2.4** What happens to the particles of matter when it contracts?
- (A) They get smaller and occupy less space.
 - (B) They get smaller and decrease in volume.
 - (C) They break the forces holding them together and spread out into a larger volume.
 - (D) They move more slowly, collide less violently, and move closer together.
- 1.1.2.5** Which is an example of diffusion?
- (A) Spilt water spreading over the floor.
 - (B) Noticing the perfume your friend is wearing.
 - (C) Molten lava flowing out of an active volcano.
 - (D) Writing with a felt pen.
- 1.1.2.6** Between which particles would you expect the forces to be weakest?
- (A) Between air particles.
 - (B) Between water particles.
 - (C) Between ice particles.
 - (D) Between lemonade particles.
- 1.1.2.7** Considering what you know about roll on deodorants, which statement is correct?
- (A) Forces between deodorant particles and skin particles are stronger than those between deodorant particles.
 - (B) Forces between deodorant particles are stronger than those between deodorant and skin particles.
 - (C) Forces between deodorant particles are equal to those between deodorant and skin particles.
 - (D) Forces between deodorant particles are equal to those between skin particles.
- 1.1.2.8** Which particles will have the least energy?
- (A) Water liquid particles at 60°C.
 - (B) Water vapour particles at 60°C.
 - (C) Water liquid particles at 50°C.
 - (D) Water vapour particles at 50°C.
- 1.1.2.9** Why do liquids diffuse more slowly than gases?
- (A) Their particles are held together more firmly.
 - (B) Their particles are not held together.
 - (C) Their particles are larger.
 - (D) Their particles move more quickly.
- 1.1.2.10** Why can't solids be poured?
- (A) Their particles are held together more strongly than in liquids.
 - (B) Their particles are not held together.
 - (C) Their particles are held together in fixed positions.
 - (D) Their particles are as close together as they can be.

1.1.2.11 Complete the following table.

	Fact	Kinetic theory explanation
(a)	Air can be compressed.	
(b)	Liquids diffuse into each other.	
(c)	Cooking odours spread throughout the house.	
(d)	You can dry yourself with a towel.	
(e)	Liquids do not fill their containers.	
(f)	Glass objects shatter when dropped onto a hard floor.	
(g)	Concrete footpaths have black rubber strips every few metres.	
(h)	Gases expand quickly when heated.	
(i)	Solids have a constant shape.	
(j)	Liquids expand when heated.	
(k)	If a soccer ball gets a hole in it, the air leaks out.	
(l)	Large bridges sit on, and are free to move on rollers at one end.	
(m)	Your breath is 'foggy' on a very cold day.	
(n)	A balloon expands on a hot day.	



1.1.3 Temperature and the kinetic theory.

1.1.3.1 Complete the sentences in the first column of the table by matching the phrase provided with the correct alternative from the second column.

Completed sentences		Second halves (not in order)
(a)	Temperature is a measure of how fast	the faster the particles are moving.
(b)	Temperature is a measure of	solid particles at the same temperature.
(c)	The higher the temperature,	the average kinetic energy of the particles of matter.
(d)	When matter is heated its	its particles lose energy and slow down.
(e)	When matter is cooled	the extra energy they have because the particles move more freely.
(f)	Gas particles have more energy due to	the particles of matter are moving.
(g)	Similarly, liquid particles have more energy than	their state than liquid particles at the same temperature.
(h)	In both cases (f) and (g), this is due to	particles absorb energy and move faster.

1.1.3.2 In terms of the kinetic theory, identify the values of and justify the concept of an absolute zero temperature.

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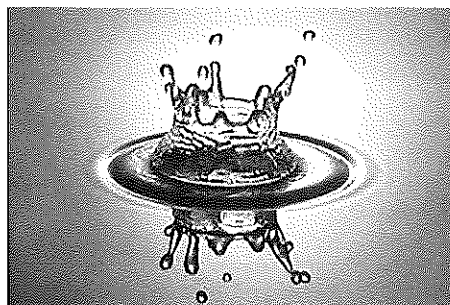
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1.1.3.3 Consider the following five pictures of water.



A: Ice at 0°C



B: Water at room temperature



C: Water at 0°C

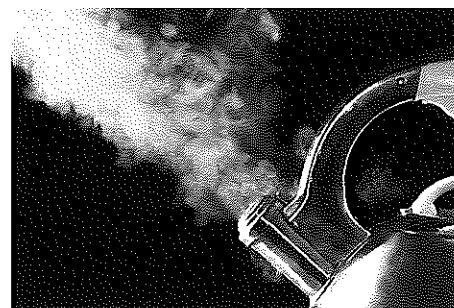
- (a) Rank the speed of the particles of water/ice/steam in order of moving fastest to slowest. Justify your answer.

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D: Steam at 100°C

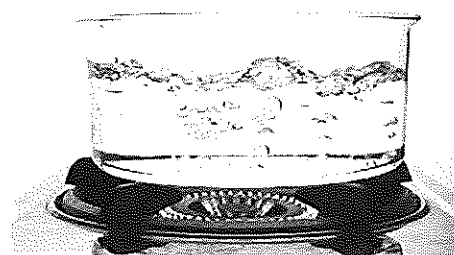
- (b) Rank the kinetic energy of the particles of water/ice/steam in order most to least. Justify your answer.

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E: Water at 100°C

- (c) In which will the water particles have the most total energy? Justify your answer.

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- (d) In which will the water particles have the least total energy? Justify your answer.

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- (e) Steam at 100°C will actually cause more severe burns to a person than water at 100°C. Hypothesise why this is so.

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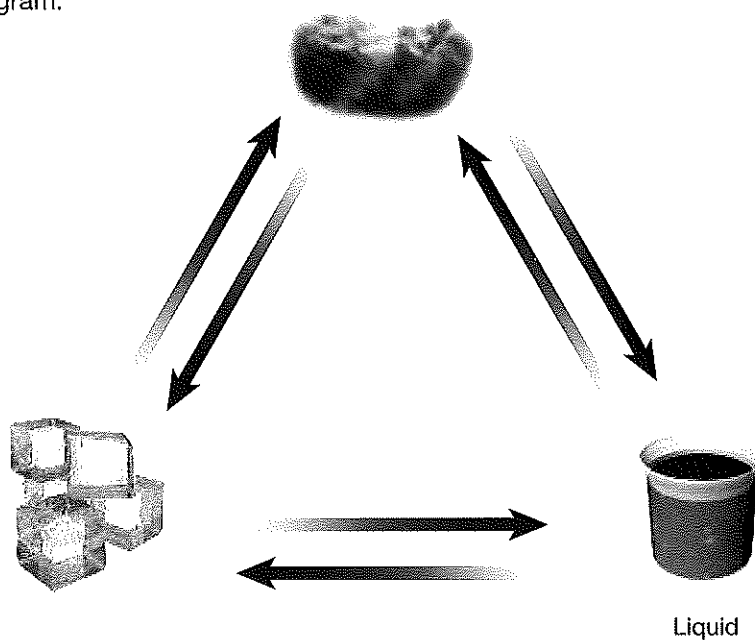
1.1.4 Changes of state and the kinetic theory.

1.1.4.1 Complete the sentences in the first column of the table by matching the phrase provided with the correct alternative from the second column.

Completed sentences		Second halves (not in order)
(a)	As a gas (or any matter) is cooled,	particles is strong enough to hold them close together.
(b)	As the particles lose energy the forces between	all particles to become bonded into the lattice.
(c)	Eventually, the strength of the forces between the	particles move into the lattice positions.
(d)	They start to form a lattice but are	condensed and become a liquid.
(e)	As more energy is removed from the system, more	them can start to hold them together.
(f)	Eventually the system loses sufficient energy for	still able to roll over one another.
(g)	The gas has	its particles move more slowly.

1.1.4.2 Place the following labels on the diagram.

Condensation
Evaporation
Gas
Liquid
Melting
Solid
Solidification
Sublimation



1.1.4.3 The diagram represents a change of state which is occurring at a constant temperature.

(a) Identify the change of state represented by the diagram.

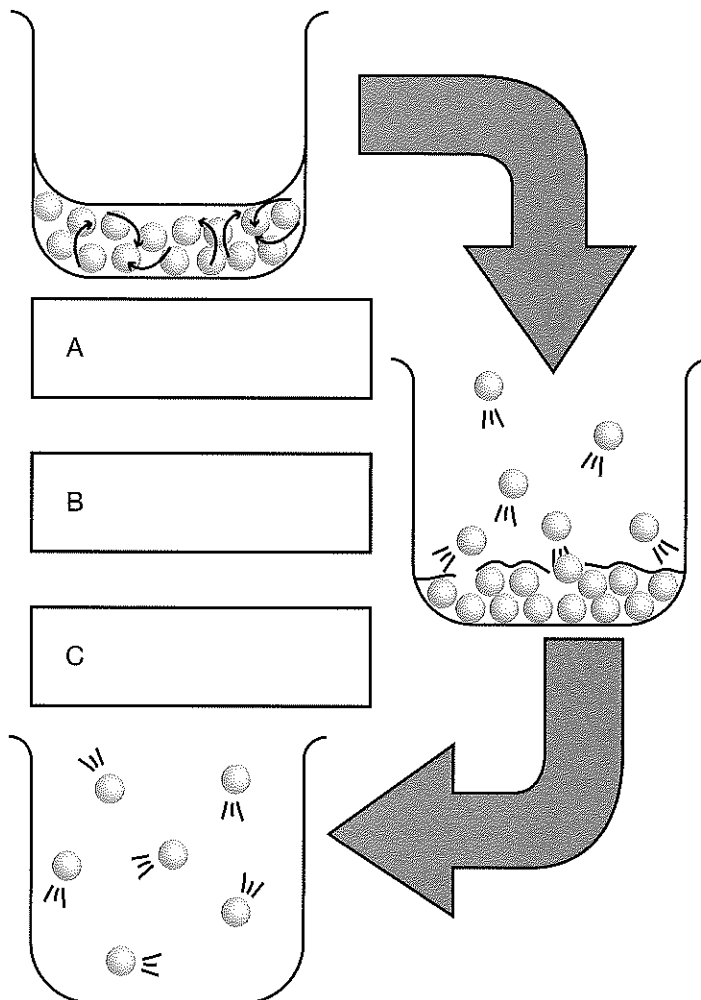
(b) What is happening to the kinetic energy of the particles as this change of state happens?

(c) What is happening to the forces between particles as this change of state happens?

(d) In which of these diagrams do the particles have the least degree of freedom of movement?

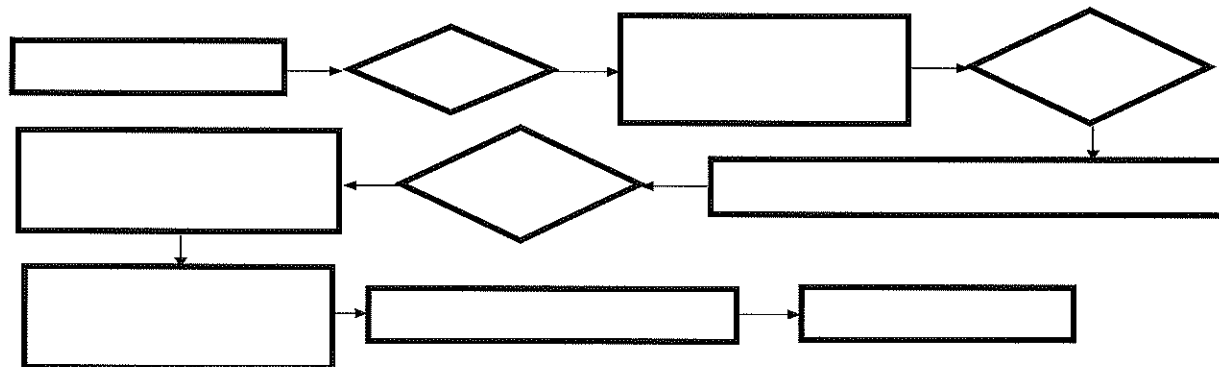
(e) Compare the rate of movement of the particles in each of the situations represented in the diagram.

(f) Write an appropriate label for each beaker (A, B and C) in the diagram.



1.1.4.4 Place the following labels in the flow chart.

Continue heating, Energy of particles increases, Energy of particles overcomes forces holding them together, Gas formed, Heat, Heat further, Liquid, Liquid evaporates, Particles break away from each other, Particles gain energy and move faster.



1.2 Temperature change, heat capacity, change of state and latent heat.

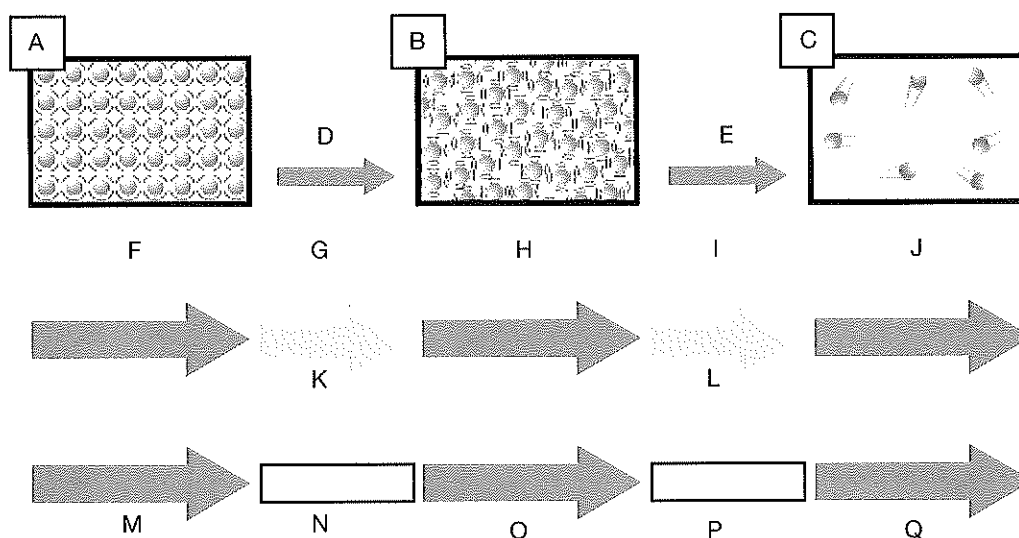
1.2.1 Changes of state and latent energy 1.

1.2.1.1

(a) What is latent energy?

(b) Explain how latent energy is used.

1.2.1.2 Consider the diagram.



Each of the labels in the table below could be placed on this diagram in place of the letters A to Q. Identify where each label belongs. Note that some labels may be used more than once.

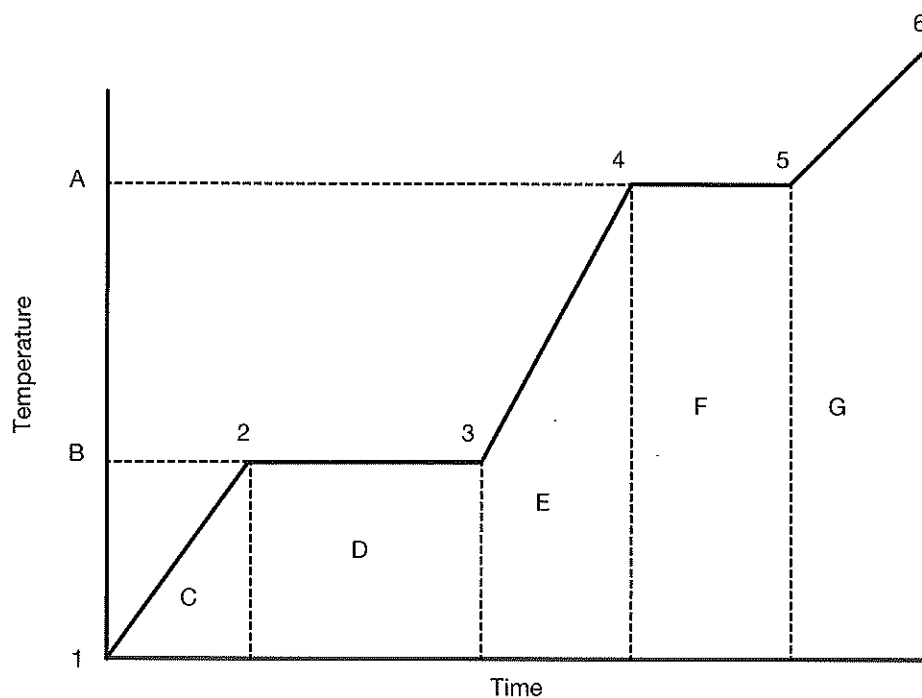
Label	Correct position(s) for label	Label	Correct position(s) for label
Boiling		Solid	
Gas		System gains kinetic energy	
Latent heat of fusion		System gains potential energy	
Latent heat of vaporisation		Temperature of system increases	
Liquid		Temperature of system remains constant	
Melting			

1.2.1.3

- (a) Explain the relationship between latent energy and the fact that in order to change the state of matter heat energy either needs to be added or taken away from it but temperature remains constant.

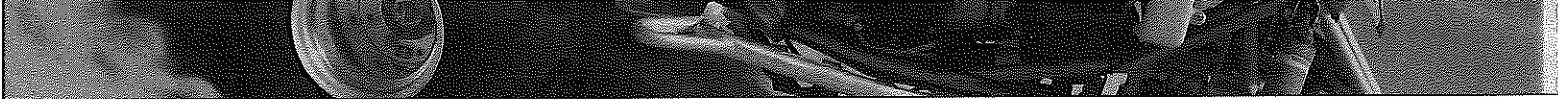
- (b) Explain why the change of state from gas to liquid is exothermic.

1.2.1.4 Consider the temperature-time graph.



- (a) Identify the states of matter present in each section of the graph as detailed in the table.

Section	State or states of matter present
C	
D	
E	
F	
G	



(b) Describe what changes (if any) are occurring in each of the following graph sections.

Section	Change occurring
1 to 2	
2 to 3	
3 to 4	
4 to 5	
5 to 6	

(c) Suggest appropriate names for the temperatures A and B.

A =

B =

(d) In which section(s) of the graph is latent heat involved? Identify the type of latent heat.

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(e) The kinetic theory diagrams below were drawn by a student doing this experiment. State what each diagram represents and match each diagram with the correct section of the graph, C to G.

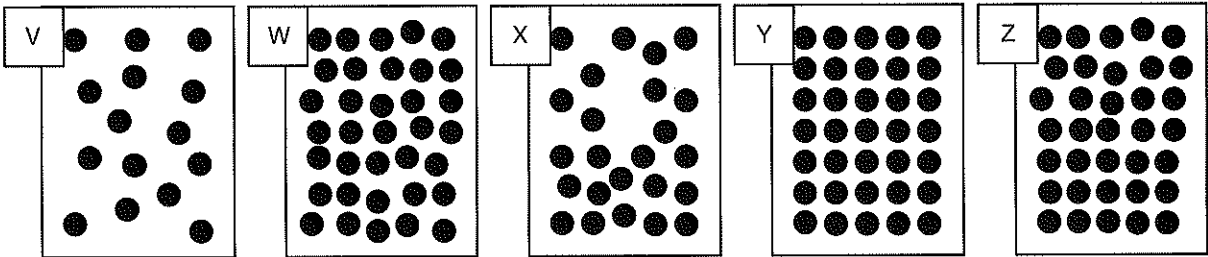


Diagram	Represents	Refers to graph section
V		
W		
X		
Y		
Z		

(f) Explain the flaw(s) in the student's diagram(s).

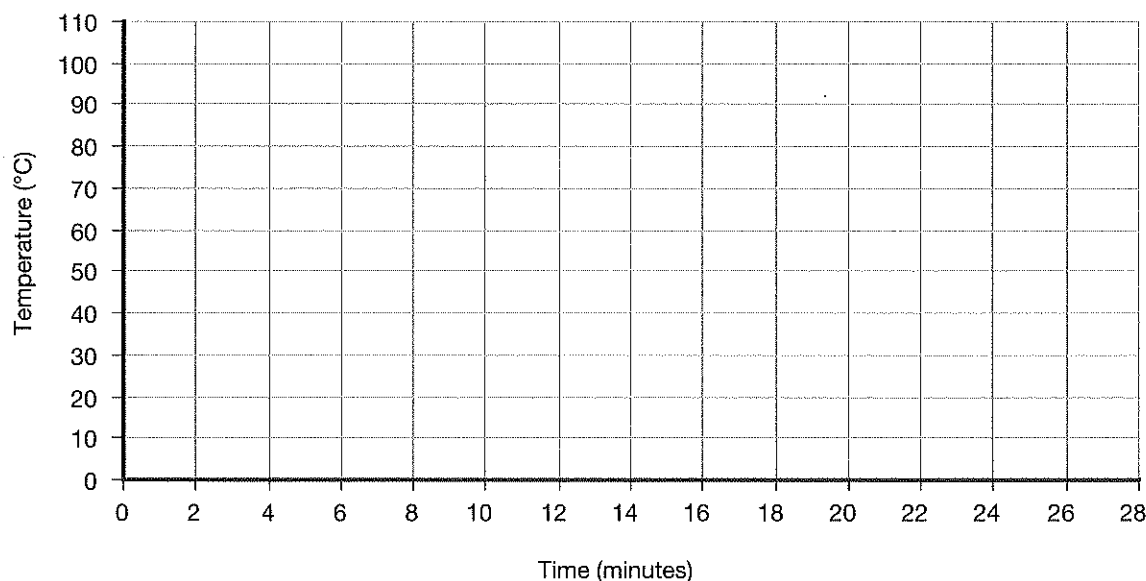
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1.2.2 Analysing a temperature change experiment.

The table shows the results of an experiment done by students who heated 200 g of ice gently until it turned into water and the water boiled and evaporated away. All water had evaporated by time 26 minutes. Use the data to answer the questions which follow it.

Time (min)	Temp (°C)	Time (min)	Temp (°C)	Time (min)	Temp (°C)
0	0	9	5	18	76
1	0	10	9	19	87
2	0	11	14	20	94
3	0	12	20	21	98
4	0	13	27	22	100
5	0	14	36	23	100
6	0	15	46	24	100
7	0.5	16	57	25	100
8	2	17	68	26	100

1.2.2.1 Graph the results of the experiment on the axes below.



1.2.2.2 Place the following labels on your graph in the appropriate positions.

Ice melting, Temperature constant, All ice has melted, Water heated, Temperature rising, Water starts boiling, Water boiling, Temperature constant



1.2.2.3 During time 0 to time 6 minutes, the ice is being heated gently, but the temperature does not rise. Suggest a reason for this.

1.2.2.4 During time 21 to time 26 minutes, the water is being gently heated still, but the temperature does not rise. Suggest a reason for this.

1.2.2.5

(a) Define latent heat of fusion.

(b) Explain how this experiment is illustrating latent heat of fusion.

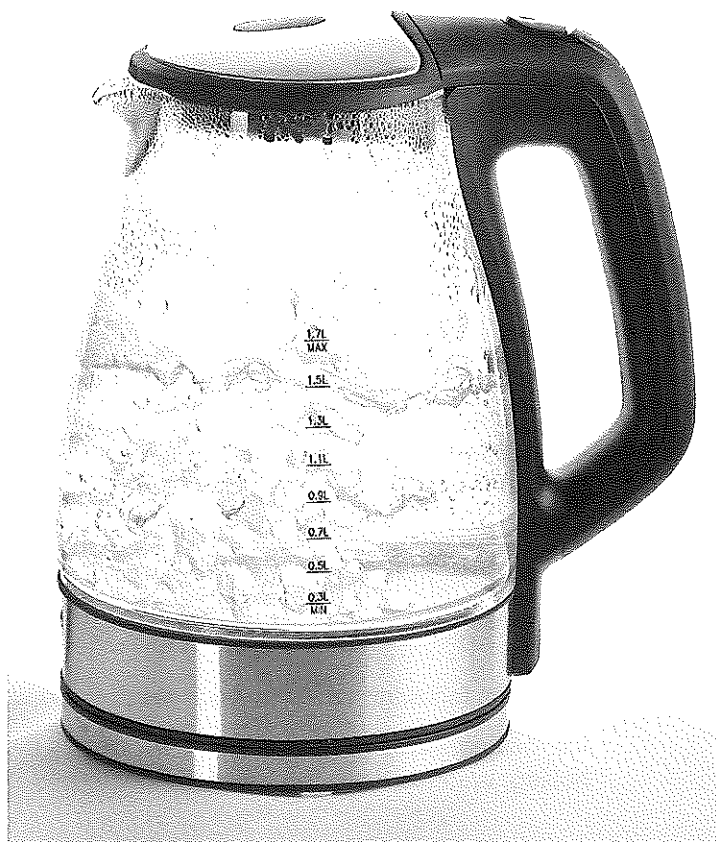
(c) While the ice is melting, its temperature does not rise until it has all melted, and yet it is still being heated. In terms of the kinetic theory, explain where the added heat energy is going.

1.2.2.6

(a) Define latent heat of vaporisation.

(b) Explain how this experiment is illustrating latent heat of vaporisation.

(c) While water is boiling, its temperature does not rise, and yet it is still being heated. In terms of the kinetic theory, explain where the added heat energy is going.



1.2.3 Changes of state and latent energy 2.

1.2.3.1

- (a) What is the relationship between latent energy and the particles of matter when a solid is melting?
.....
.....
- (b) What is the relationship between latent energy and the particles of matter when a liquid evaporates?
.....
.....
- (c) What is the relationship between latent energy and the particles of matter when a liquid freezes and becomes solid?
.....
.....
- (d) What is the relationship between latent energy and the particles of matter when a gas condenses to form a liquid?
.....
.....

1.2.3.2 The latent heats of fusion of nitrogen and alcohol are 26 kJ kg^{-1} and 104 kJ kg^{-1} respectively. A certain mass of nitrogen releases $X \text{ kJ}$ of energy when it freezes.

- (a) What mass of the alcohol will release the same amount of energy when it freezes?
- (b) How much energy will the same mass of alcohol release when it freezes?
- (c) What mass of each substance releases 15.6 kJ of energy when it freezes?

1.2.3.3 If 4.2 kJ of energy is required to evaporate 20 g of ethanol, find the specific heat of vaporisation of ethanol.
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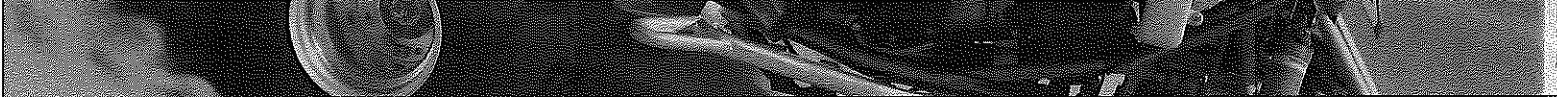
1.2.3.4 2.5 kg of gold in a 12 kW furnace has reached its melting point and is about to start melting. If the latent heat of fusion of gold is 64.4 kJ kg^{-1} :

- (a) Find the energy required to melt all the gold.
- (b) Calculate how long this will take.

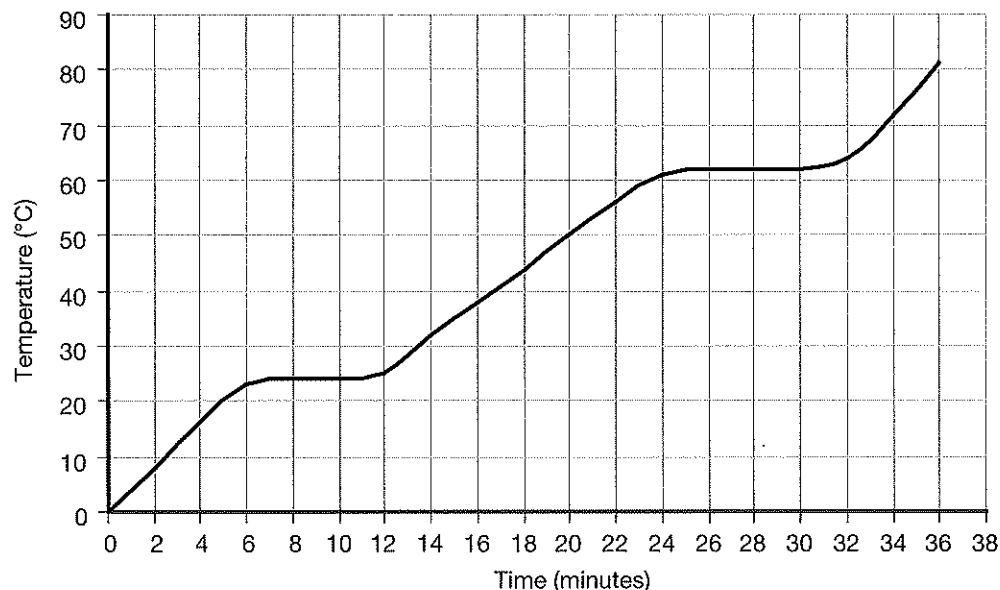
1.2.3.5 An iceblock tray in a refrigerator holds 125 mL of water. The ice was removed from the freezer and put in an empty jug. 40 minutes later the ice had all melted. If the latent heat of fusion of ice is 333 kJ kg^{-1} find how much energy was released into the air as the ice melted.
.....

1.2.3.6 Energy is supplied at a uniform rate to 250 g of lead at its melting point. It melts totally in exactly 10.0 minutes. If the latent heat of fusion of lead is 24.5 kJ kg^{-1} find:

- (a) The total energy required to melt the lead.
- (b) The rate at which the energy is supplied to the lead.
- (c) The energy needed to melt 250 g of copper if its latent heat of fusion is $1.34 \times 10^2 \text{ kJ kg}^{-1}$



- 1.2.3.7** The graph shows the temperature-time curve for a 45 g solid which was placed in a well insulated, massless container and heated uniformly by a 500 W heater.



- (a) What is the melting point of the solid?
- (b) What is its boiling point?
- (c) If the heater was a 240 V heater, what current did it deliver to the heating coil?
- (d) What was the total energy supplied by the heater from the start of the experiment until the last of the water was evaporated?
.....
.....
- (e) Without making any calculations, predict which is larger, the latent heat of fusion, or the latent heat of vaporisation of the substance. Justify your prediction.
.....
.....
.....
- (f) Calculate the latent heat of fusion of the substance.
- (g) Calculate the latent heat of vaporisation of the substance.
- (h) Calculate the specific heat of the liquid substance.

1.2.4 Transferring heat energy 1.

1.2.4.1 Complete the sentences in the first column of the table by matching the phrase provided with the correct alternative from the second column.

Completed sentences		Second halves (not in order)
(a)	Different substances are made up	which it moves ($E_k = \frac{1}{2}mv^2$).
(b)	Different atoms have	the same amount of particles with smaller masses.
(c)	Kinetic energy of a particle is due to its mass and the rate at	compared to increasing the temperature of 10 kg of substance B by the same amount.
(d)	The temperature of matter is a measure of the average	different masses.
(e)	Particles with greater mass will require more energy to increase their kinetic energy than	of the same material by the same amount.
(f)	So raising the temperature of 10 kg of substance A by $T^\circ\text{C}$ will require a different amount of energy	of different combinations of atoms.
(g)	In addition, raising the temperature of 10 kg of a substance by $T^\circ\text{C}$ will require more energy than raising the temperature of 5 kg	kinetic energy of its particles.

1.2.4.2 Use the information in your completed sentences from Question 1.2.4.1 to complete the following which is a generalised conclusion based on these sentences.

The amount of energy required to increase or decrease the temperature of a substance depends on

- 1.2.4.3** Complete the sentences in the first column of the table by matching the phrase provided with the correct alternative from the second column.

Completed sentences		Second halves (not in order)
(a)	Different substances are made up of different	the rate at which it moves ($E_k = \frac{1}{2}mv^2$).
(b)	Different atoms and molecules within substances are held together	measure of the average kinetic energy of its particles.
(c)	Kinetic energy of a particle is due to its mass and	by forces that vary in strength.
(d)	The temperature of matter is a	as particles which are held together with weaker force.
(e)	Particles held together with larger forces will require more energy to increase their kinetic energy the same amount	combinations of atoms.

- 1.2.4.4** Use the information in your completed sentences from Question 1.2.4.3 to complete the following which is a generalised conclusion based on these sentences.

The amount of energy required to increase or decrease the temperature of a substance depends on

- 1.2.4.5** Complete the following.

The amount of energy required to increase or decrease the temperature of a substance also depends on how far

- 1.2.4.6** On the basis of your answers to the questions above, give a definition for the specific heat of a substance.

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Use the following information to answer the next FOUR questions.

Substance	Latent heat of fusion (kJ kg ⁻¹)	Latent heat of vaporisation (kJ kg ⁻¹)	Specific heat (kJ kg ⁻¹ °C ⁻¹)
Ammonia	33	137	2.09
Copper	134	5060	0.39
Ethanol	104	850	2.44
Glass			0.84
Gold	64.4	1580	0.13
Iron	289	6340	0.45
Silver	88	2300	0.23
Water	333	2260	4.18

1.2.4.7

- (a) Predict the rank of ammonia, alcohol and water in order from lowest to highest boiling points. Justify your answer.
-
- (b) Which substance, gold, iron or silver would have the lowest melting point? Justify your answer.
-
- (c) If 50 g samples of each of the substances in the table has X kJ of energy added to it, which would rise in temperature by the greatest amount? Justify your answer.
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1.2.4.8 Calculate the energy required to melt 250 g of the following.

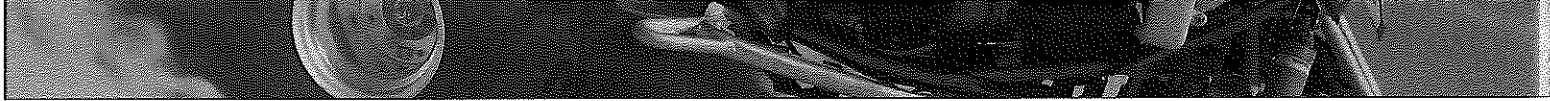
- (a) Copper.
- (b) Gold.
- (c) Iron.
- (d) Water.

1.2.4.9 Calculate the energy required to raise the temperature of each of the following substances from 18°C to 150°C.

- (a) 150 g ammonia.
- (b) 250 g glass.
- (c) 1.5 kg iron.
- (d) 2500 kg silver.

1.2.4.10 Calculate the total energy required to do each of the following.

- (a) Melt 75 g of ice then boil the water.
- (b) Evaporate 375 g of ice.
- (c) Freeze 500 g of water vapour at 100°C.



1.3 Heat transfer by conduction, convection and/or radiation.

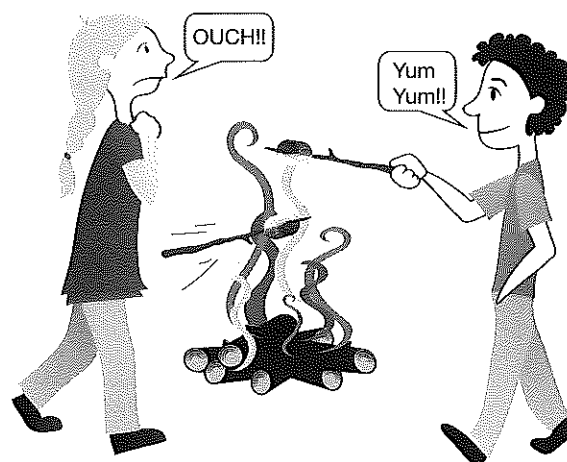
1.3.1 Heat conduction.

- 1.3.1.1 Most of the sentences below have errors in them. Either one or two words need to be changed/added/removed to correct each sentence. There may be the occasional correct sentence which does not need any changes. Your task is to rewrite the sentences, correcting those that have errors, so that what they say is correct to summarise basic ideas about the conduction of heat energy.

Incorrect sentences	Correct sentences
(a) Metals and metal alloys don't allow heat energy to transfer through them.	
(b) For this reason they are known as heat insulators.	
(c) Non-metallic substances allow heat energy to pass through them.	
(d) These substances are called heat conductors.	
(e) Conduction of heat occurs because any applied energy causes the particles of the metal to vibrate more slowly.	
(f) This decreased vibration causes these particles to bump into the particles next to them and cause these to vibrate less as well.	
(g) Gradually the heat energy is passed through the metal from particle to particle.	
(h) In addition to resulting in conduction of heat, the decreased vibrations cause the metal to expand.	
(i) This happens because the particles need less space in which to vibrate.	
(j) Air is one of the best conductors.	
(k) Many substances that contain a lot of air, such as cork, polystyrene foam, fibreglass and wool, are therefore good conductors.	

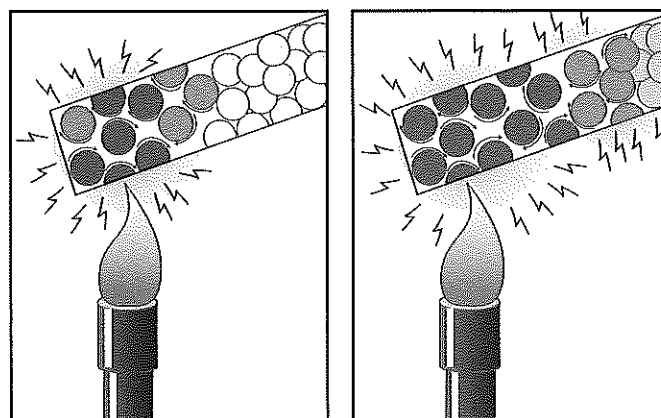
- 1.3.1.2 The cartoon shows two people heating marshmallows on rods.

Explain the comments by each person.



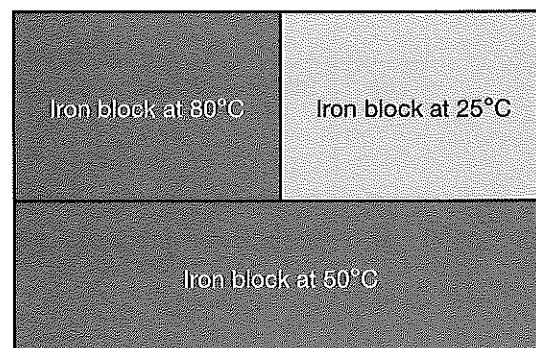
- 1.3.1.3 The diagrams show a rod being heated by a Bunsen burner flame.

Student X claims that the rod is metallic, student Y claims it is probably a glass rod. Which student is correct? Justify your answer.

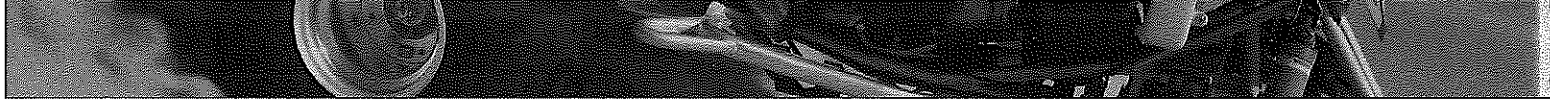


- 1.3.1.4 Consider the following diagram.

- (a) On the diagram draw in arrows to indicate the direction of energy flow between these blocks.
- (b) What will happen to these blocks if they are left to stand for an indefinite time?



- (c) Estimate the final temperature of the three blocks explaining how you arrive at your estimate and any assumptions you make.



1.3.2 Heat convection.

1.3.2.1 Complete the sentences in the first column of the table by matching the phrase provided with the correct alternative from the second column.

Completed sentences		Second halves (not in order)
(a)	As water at the bottom of a beaker is heated,	to the top of the beaker.
(b)	As the water expands	convection occurs in gases and liquids.
(c)	The less dense water then 'floats'	pushes the cooling water to the sides.
(d)	At the top of the beaker this water	it expands.
(e)	More rising hot water from the bottom	flow, unlike solids which maintain a constant shape.
(f)	The cooling water, now more dense, sinks to the bottom	cools down as it is further from the heat source.
(g)	This occurs because the particles in the water	more freely than the particles in solids.
(h)	The transfer of heat energy by	it becomes less dense.
(i)	Gases and liquids are collectively known as fluids because they both	are not held together tightly but can flow over one another.
(j)	This is also because their particles can move	of the beaker to take the place of more rising hot water.

1.3.2.2 Which of the two animals prefers warmer conditions? Justify your answer.

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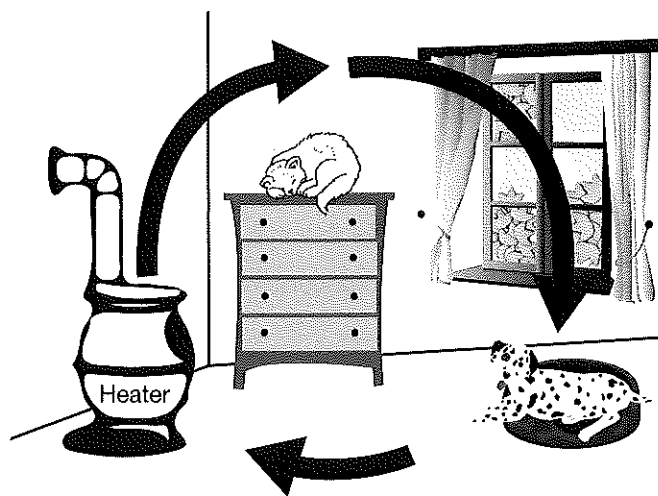
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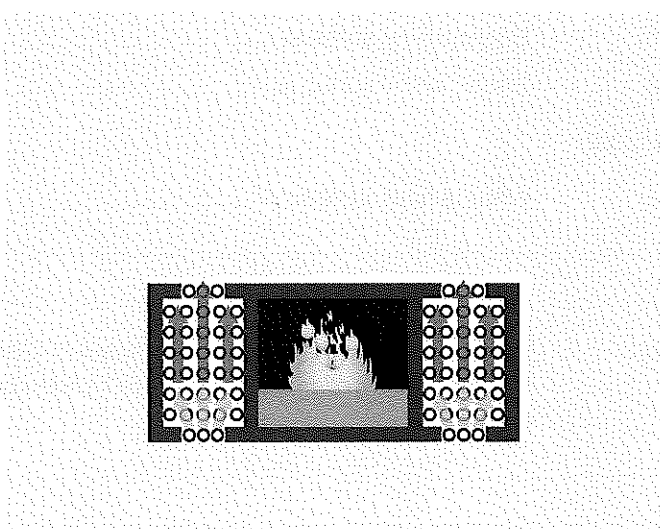


1.3.2.3 The diagram represents a space heater in a room.

(a) Draw in two convection currents to show how this space heater works to warm the air.

On your diagram label the following.

- (b) What air will be the coolest. (A)
- (c) Where air will be the least dense. (B)
- (d) Where warm air is rising. (C)
- (e) Where cooler air is falling. (D)



1.3.2.4 Gases transfer heat energy by convection better than liquids and solids do not transfer heat energy by convection. In terms of the particle theory of matter, explain these facts.

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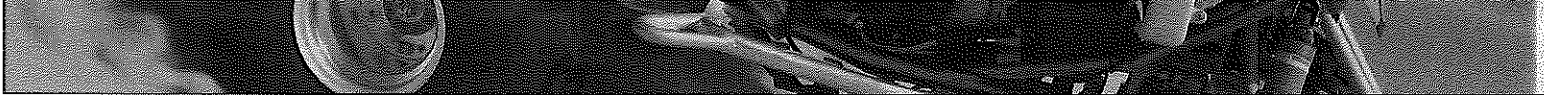
1.3.2.5 Some swimming pools are covered during summer when not in use by a black cover which absorbs radiant heat energy from the Sun. Evaluate the role of convection in the subsequent heating of the water in the pool.

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1.3.3 Heat radiation.

1.3.3.1 Complete the following sentences by inserting the missing word(s).

- (a) Heat energy can also travel through fluids and space by a process known as
- (b) Radiation does involve the particles of matter.
- (c) It is because of this that heat energy can be through space or a vacuum.
- (d) Heat radiation occurs through waves, also known as rays.
- (e) When infra-red radiation hits a surface it is partly and partly
- (f) It is the energy absorbed by a substance that causes its to rise.
- (g) Dull, dark surfaces absorb heat energy more than bright, shiny surfaces.
- (h) Bright, shiny surfaces heat energy more than dull, dark surfaces.
- (i) Bright, shiny surfaces heat energy more rapidly than dull, black surfaces.

1.3.3.2

- (a) What is heat radiation?
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- (b) When you put your hands at the sides of a Bunsen burner flame you feel very little heat, if any, from the flame. However, if you put your hands above the flame (please don't) you can feel much more heat energy. Why the difference?
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- (c) What are the two factors that affect the rate at which an object will both absorb or emit heat energy?
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1.4 Transferring energy and thermal equilibrium.

1.4.1 Transferring heat energy 2.

Show all working and logic in each question so it is easier to follow what you have done.

- 1.4.1.1** 250 g of water, specific heat $4.18 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$, at 80°C is added to a 50 g copper calorimeter at the same temperature, and 40 g crushed ice at 0°C is added. If the system is properly insulated, find the final temperature of the calorimeter and the water. Take the specific heat of copper as $0.39 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ and the latent heat of fusion of ice as 333 kJ kg^{-1} .

- 1.4.1.2** In order to warm a 200 mL glass of cold water, initially at 10°C , 50 mL of water at 80°C was added to it.

(a) Ignoring any heat energy transferred to the glass, what would be the final temperature of the water?

(b) If the glass had a mass of 250 g, and was initially at the same temperature as the cold water, what would be the final temperature of the water and glass? Take the specific heat of glass as $0.84 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

- 1.4.1.3** To both dilute and cool a concentrated ammonia solution, 130 g of ice is added to the concentrated ammonia at 24°C in an insulated calorimeter. Ignoring any energy transfer to or from the calorimeter, if the final temperature of the diluted ammonia is 15°C , find the initial volume of concentrated ammonia used. Take the specific heat of ammonia as $2.09 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$, and the latent heat of fusion of ice as $333 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.



1.4.1.4 50 g of water at 75°C is mixed with 40 g of water at 15°C in an copper beaker which has a mass of 150 g, and is at 20°C. The specific heat of copper is 0.39 kJ kg⁻¹ °C⁻¹.

(a) Ignoring any heat transferred to or from the beaker, find the final temperature of the water.

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(b) Considering heat transferred to or from the beaker, find the final temperature of the water.

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1.4.1.5 A 2 kg block of metal iron, at 15°C is placed on top of a 3 kg block of copper at 125°C. If the blocks are in a perfectly insulated environment, what will be their final temperature? The specific heats of iron and copper are 0.45 and 0.39 kJ kg⁻¹ °C⁻¹ respectively.

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1.4.1.6 Copper has a specific heat of 0.39 kJ kg⁻¹ °C⁻¹ and water has a specific heat of 4.18 kJ kg⁻¹ °C⁻¹. What will be the final temperature of the contents in an insulated container if 180 g of copper at 90°C is placed into 250 mL of water at 15°C?

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1.4.1.7 9200 J of heat energy was used to raise the temperature of 300 g of a metal from 20°C to 45°C. Find the specific heat of the metal.

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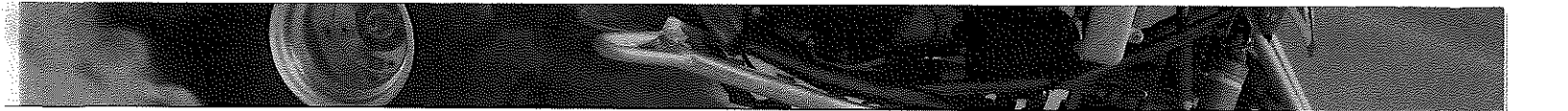
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1.4.1.8 A 90 g piece of metal is heated from 20°C to 180°C. If the specific heat of the metal is 0.45 kJ kg⁻¹ °C⁻¹, how much heat energy has been absorbed by the metal?

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1.5 Thermal energy, mechanical work, energy transfers and transformations.

1.5.1 Work done and internal heat energy.

- 1.5.1.1** A heated gas expands, raising a piston. Which of the following describes the energy exchanges of this process?
- (A) Energy is transferred from the heat source to the piston then to the gas.
 - (B) Energy is transferred from the heat source to the gas then to the piston.
 - (C) Energy is transferred to the gas in the form of heat and work done by the piston.
 - (D) Energy is transferred directly to the piston from the heat source.
- 1.5.1.2** A gas in a sealed cylinder is heated. Which of the following does *not* increase as the gas is heated?
- (A) The average number of gas molecules hitting the cylinder walls per second.
 - (B) The average kinetic energy of the gas molecules.
 - (C) The average speed of the gas molecules.
 - (D) The average distance between the gas molecules.
- 1.5.1.3** When a gas is heated in a closed container, the internal pressure increases. Which best describes the reason for the increased pressure?
- (A) The average kinetic energy of the gas molecules decreases.
 - (B) The potential energy of the gas increases.
 - (C) The average kinetic energy of the gas molecules increases.
 - (D) The potential energy of the gas decreases.
- 1.5.1.4** An engine has an input of heat energy of 10 750 J and does 2420 J of work. Which of the following is the heat loss?
- (A) 0.225 J
 - (B) 4.44 J
 - (C) 8330 J
 - (D) 13 170 J
- 1.5.1.5** The pressure of a gas inside a closed, rigid container will increase when the gas temperature increases. The pressure of the gas increases because the:
- (A) Density of the gas decreases.
 - (B) Rate of collisions of gas molecules with the surface increases.
 - (C) Container expands in size when heated.
 - (D) Gas molecules bond together to form more massive molecules.
- 1.5.1.6** A steel block at 250°C is placed on top of a copper block at 100°C. The thermal energy of the copper begins to increase. What is the source of this increase in energy?
- (A) The work done by the molecules within the copper.
 - (B) The work done by the interaction of the two metals.
 - (C) Heat flowing by means of conduction.
 - (D) Heat flowing by means of radiation.
- 1.5.1.7** What is the internal heat energy of an object or system?
- (A) The energy responsible for its temperature.
 - (B) The energy responsible for its internal pressure.
 - (C) The energy bound up in the bonds between its constituent particles.
 - (D) The energy due to its state and temperature.



- 1.5.1.8** What happens to the internal thermal energy of a gas in a sealed container when it is compressed?
- (A) It increases because work is done on it.
 - (B) It decreases because it does work against the compressive force.
 - (C) It remains constant.
 - (D) It increases because it does more work on its container walls.
- 1.5.1.9** How is the internal energy of a system increased?
- (A) Adding heat to the system.
 - (B) Having the system do work on the surroundings.
 - (C) Withdrawing heat from the system.
 - (D) Adding heat to the system and having the system do work on the surroundings.
- 1.5.1.10** Which one of the following conditions would always result in an increase in the internal energy of a system?
- (A) The system loses heat and does work on the surroundings.
 - (B) The system gains heat and does work on the surroundings.
 - (C) The system loses heat and has work done on it by the surroundings.
 - (D) The system gains heat and has work done on it by the surroundings.
- 1.5.1.11** What is the value of ΔE for a system that performs 111 kJ of work on its surroundings while gaining 89 kJ of heat?
- (A) 200 kJ
 - (B) -200 kJ
 - (C) -111 kJ
 - (D) -22 kJ
- 1.5.1.12** A gas is kept at a constant pressure of 70.0 kPa while its volume decreases by 0.20 m³ at constant temperature. What is the work done by the system on its environment?
- (A) 14 kJ
 - (B) -14 kJ
 - (C) 35 kJ
 - (D) -35 kJ
- 1.5.1.13** A system contains 50 L of gas at a constant volume of 4.0 L. If 100 kJ of thermal energy is transferred to the system, what is the change in the internal energy of the system?
- (A) 0
 - (B) 100 kJ
 - (C) -100 kJ
 - (D) 400 kJ
- 1.5.1.14** Which of the following statements is correct?
- (A) When a system absorbs heat and does work, the change in its internal energy is always negative.
 - (B) When a system emits heat and does work, the change in its internal energy is always negative.
 - (C) When a system emits heat and has work done on it, the change in its internal energy is always negative.
 - (D) When a system absorbs heat and has work done on it, the change in its internal energy is always negative.

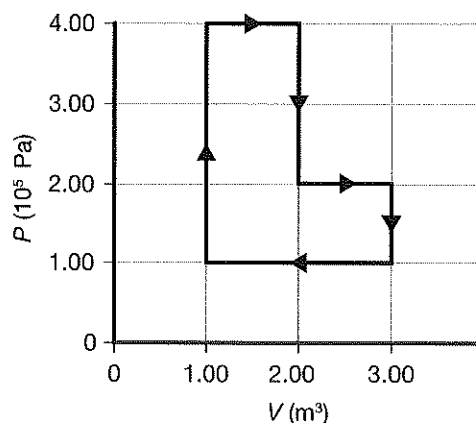
- 1.5.1.15** Some students made statements about how the internal energy of a system could be increased.
 Student X: By transferring heat from the surroundings to the system.
 Student Y: By transferring heat from the system to the surroundings.
 Student Z: By doing work on the system.

Which student(s) are correct?

- (A) X only. (B) Y only. (C) Both X and Z. (D) Both Y and Z.

- 1.5.1.16** How much net work is done by the gas undergoing the cyclic process illustrated in the diagram? Choose the best estimate.

- (A) 3×10^5 J
 (B) 4×10^5 J
 (C) 5×10^5 J
 (D) 6×10^5 J

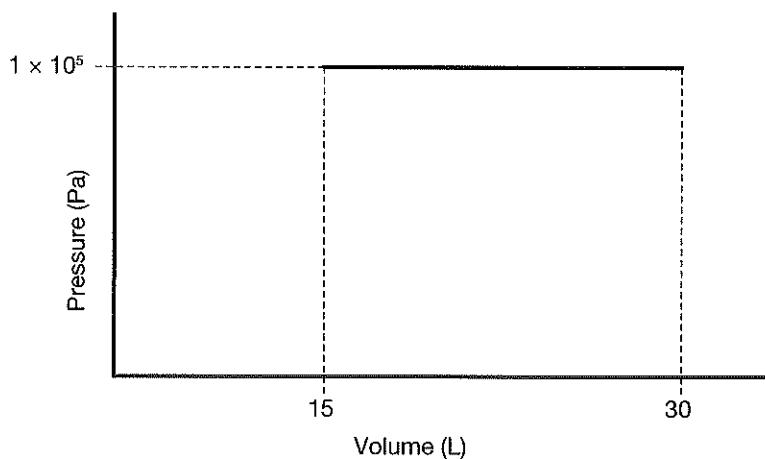


- 1.5.1.17** A balloon filled with gas expands at constant pressure of 6×10^5 kPa from a volume of 4.0 L to a volume of 6.0 L. How much work is done by the gas?

- 1.5.1.18** In an experiment, a 15 L sample of a gas is initially at 80°C and 1.2×10^5 pascals pressure, and is contained in a cylinder with a freely moving piston.

It was heated at constant pressure until its volume had doubled.

The graph represents this experiment.



- (a) What is represented by the area under the graph?

- (b) How much work is done by the gas in expanding?

- (c) How much work would have to be done to compress it back to its original volume?

- (d) Predict what happens to the temperature of the gas as it firstly expands to double the volume, and is then compressed back to its original volume. Explain your answer.



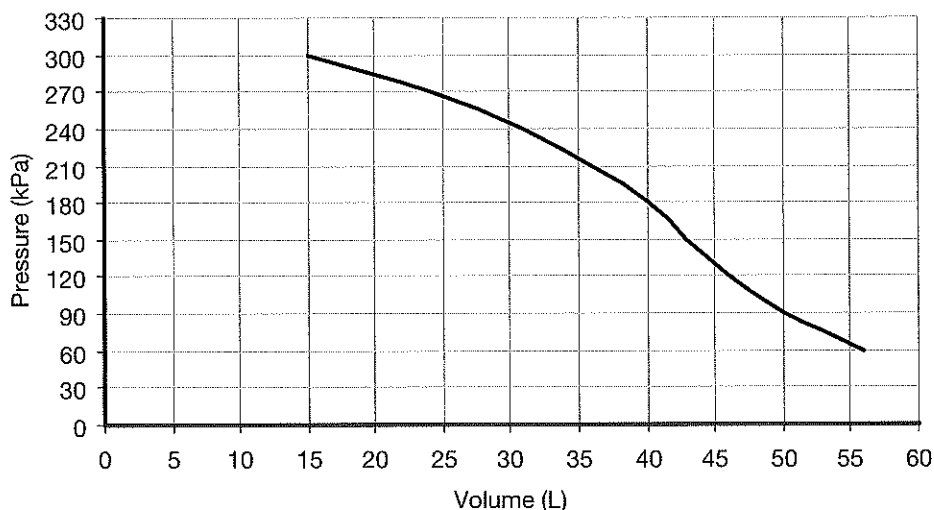
1.5.1.19 A gas in a container fitted with a movable piston is heated. Its volume increases so that its pressure remains constant at 180 kPa.

- (a) Is work done on this system? If so, by what?
- (b) What happens to the internal thermal energy of the system?

The gas is then cooled back to its original temperature again at constant pressure. During this step, its volume changes from 5.3 L to 2.8 L.

- (c) Find the change in the internal thermal energy of the gas in this process.
- (d) How much work would have been done on the piston in (a)?

1.5.1.20 The graph shows how the volume of 250 g of gas with a heat capacity of $0.6 \text{ kJ kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ changed as the pressure acting on it was increased.



- (a) Is work done on or by the gas in this situation? Justify your answer.
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- (b) How much work was done?
- (c) If all the work done was converted into thermal energy, by how much did the temperature of the gas rise?
.....

1.5.1.21 1.80 L of air is heated in a container with a piston which allows it to expand and maintain a constant pressure of 135 kPa. The final volume of the gas was 2.10 L.

- (a) How much work was done on the air?
- (b) Given that the pressure remains constant, would the temperature of the gas increase, decrease or stay the same? Justify your answer.
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1.6 Science as a global enterprise, developing models and theories.

1.6.1 Development of thermodynamics.

- 1.6.1.1 The two tables show the names of several scientists involved in the development of thermodynamics and some of the work attributed to each. The work is listed in chronological order, but the scientists are not. Your task is to match the scientists to their work.

Contribution to thermodynamics	Correct order of scientists	Scientists (alphabetical)
In 1698, he developed and patented the first crude steam engine to pump water out of coal mines.		Joseph Black (1728-1799)
Invented the first atmospheric steam engine, first used in 1712 to pump water out of a mine.		Nicolas Carnot (1796-1832)
In the early 1820s, hypothesised the existence of an engine which ran on a reversible cycle. His ideas led to the development of the second law of thermodynamics.		Josiah Gibbs (1839-1903)
Known for his improvements of the steam engine. In 1765, his work improving the Newcomen engine helped bring about the Industrial Revolution.		James Joule (1818-1889)
Did experiments on the freezing and melting of liquids that led to the concept of latent heats of fusion and vaporisation. His work led to the concepts of heat capacity or specific heat.		Antoine Lavoisier (1743-1794)
Helped explain what happened when substances were burned. In 1783 he designed the calorimeter for measuring the amount of heat given off during combustion or respiration.		James Clerk Maxwell (1831-1879)
Connected mechanical and heat energy. In 1850 he published a calculated value for the <i>mechanical equivalent of heat</i> which is defined in an early version of the first law of thermodynamics.		Thomas Newcomen (1663-1729)
Defined the absolute temperature scale in 1847. In 1851 he published ideas leading to the second law of thermodynamics and supported his friend James Joule's mechanical equivalent of heat. Introduced the term 'kinetic energy' in 1856.		William John Rankine (1820-1872)
Established relationships between the temperature, pressure and density of gases, and expressions for the latent heat of evaporation of a liquid. Formulated a science of energetics describing dynamics in terms of energy and its transformations rather than force and motion.		Thomas Savery (1650-1715)
One of his most important achievements was his contribution to kinetic theory of gases. Calculated the mean free path of molecules and formulated the equations for the kinetic theory of gases.		William Thomson (Lord Kelvin) (1824-1907)
Created statistical mechanics explaining the laws of thermodynamics. He published his ideas in 1875 and 1878 in a work which consisted of about three hundred pages and contains exactly seven hundred numbered mathematical equations.		James Watt (1736-1819)

1.6.2 The internal combustion engine.

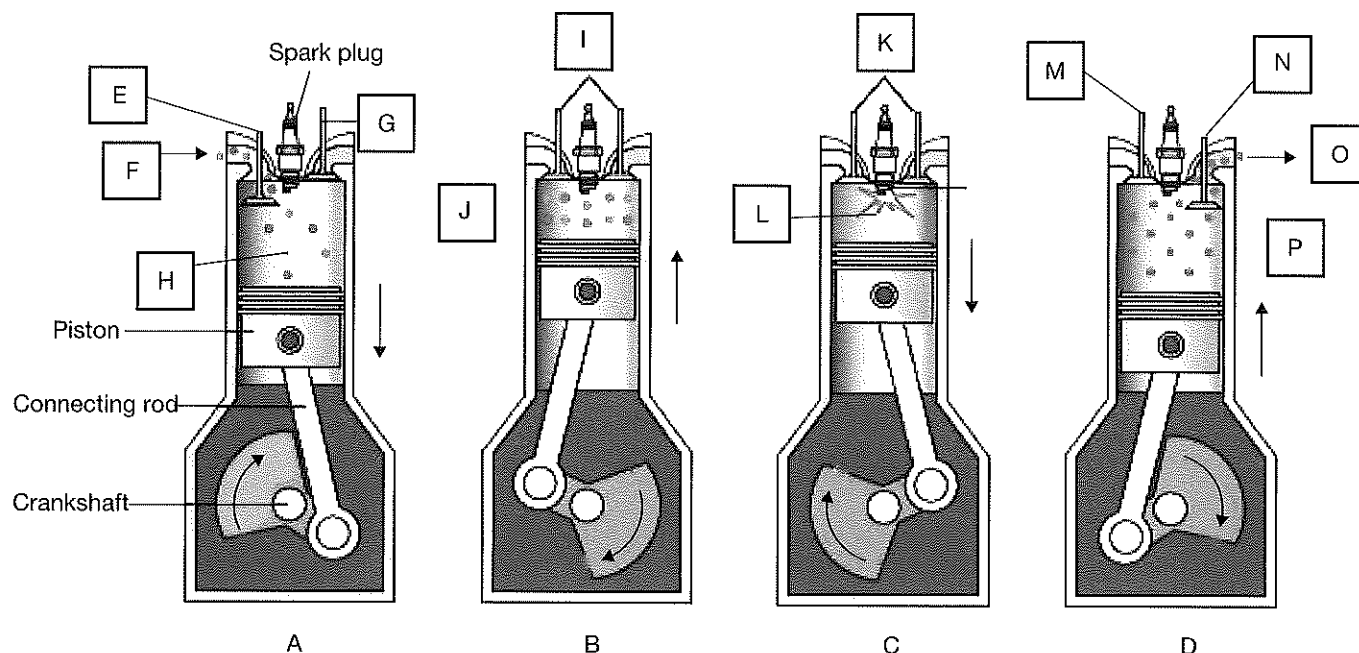
1.6.2.1 Complete the sentences to show how the conversion process of the chemical energy in a fuel into mechanical energy by an internal combustion engine can be explained.

	Completed sentence	Second halves (not in order)
(a)	Chemical energy of fuel is converted to thermal energy by	pressure of the gases within the engine cylinder.
(b)	Thermal energy raises the temperature and	the resistance of the engine mechanism.
(c)	High pressure gases expand against	oxidation with air in the combustion chamber.
(d)	The expansion of the gases moves the	piston up ready for the next cycle of the engine.
(e)	Movement of the piston up and down produces	a rotational torque which turns the crankshaft.

1.6.2.2 The four stroke internal combustion engine is so called because it has four different steps in each cycle of its operation. These are listed in order of operation in the first column of the table below.

- (a) Describe, in the second column of the table, what happens during each stroke.
(b) Copy the letter of the diagrams, of an engine (A to D) which matches each stroke into column three.

Stroke	What happens during this stroke	Diagram
Intake stroke		
Compression stroke		
Combustion stroke (also called the power stroke)		
Exhaust stroke		



1.6.2.3 The table on the left shows the 12 labels on the four diagrams. The table on the right shows possible labels for each of these labels. Your task is to select the correct label from the list of possibles and write it against the letter from each position on the diagrams in Table 1. Note that some labels may be used more than once.

Table 1		Table 2
Position	Correct label	Possible labels
E		Air-fuel mixture
F		Combustion chamber
G		Compressed air-fuel mixture
H		Exhaust gases
I		Exhaust valve closed
J		Exhaust valve open
K		Intake valve closed
L		Intake valve open
M		Spark plug firing
N		Valves closed
O		
P		

1.7 The use of scientific knowledge.

1.7.1 The energy balance of Earth.

1.7.1.1 Complete the sentences in the first column of the table by matching the phrase provided with the correct alternative from the second column.

Completed sentences		Second halves (not in order)
(a)	100% of the energy entering Earth's	atmosphere.
(b)	About 50% of the energy from the Sun is absorbed	space and Earth's energy balance is fairly well maintained.
(c)	About 30% of the Sun's energy is directly reflected	atmosphere through condensation.
(d)	About 20% of the Sun's energy is absorbed by the	clouds and gases in the atmosphere.
(e)	Most of the energy re-emitted from the surface does not	by the Earth's surface, the land and oceans.
(f)	This re-emitted energy is reabsorbed by	the Earth a habitable temperature.
(g)	Some of the re-emitted	the amount of energy the planet radiates to space as heat.
(h)	More re-emitted energy is released into the	go directly out to space.
(i)	The majority of the energy re-emitted from the surface	way out is responsible for global warming.
(j)	These gases constantly emit the Sun's energy back into the atmosphere and keep	atmosphere comes from the Sun.
(k)	Eventually, most of the energy makes its way back out to	defines expectations for future climate change.
(l)	The energy that doesn't make its	energy gets redistributed by convection.
(m)	Earth's energy imbalance is the difference between the amount of solar energy absorbed by Earth and	is absorbed by the greenhouse gases.
(n)	Earth's energy imbalance is the single most crucial measure of the status of Earth's climate and it	back to space by the atmosphere.

1.7.1.2 One idea about global warming is the following.

If the atmosphere gets warmer, more energy will be absorbed from the atmosphere by surface water and more water will evaporate into the atmosphere. Here it will condense, releasing the energy back into the atmosphere where it can be radiated back into space, thus decreasing the extent of the global warming, or at least maintaining the balance.

Comment on the scientific logic in this idea.

1.7.1.3 Consider this statement.

'The lack of precise knowledge of surface energy changes profoundly affects our ability to understand how Earth's climate responds to increasing concentrations of greenhouse gases.'

'For example, latest satellite observations of global precipitation indicates that more precipitation is occurring than previously thought. This additional precipitation is driven by more energy leaving the surface by evaporation – that is, in the form of absorbed latent heat of evaporation. This may offset much of the increase in solar energy absorbed at the surface.'

'This example is only one of many which shows that all current climate models are based on bad assumptions. And because the raw output of those models do not reproduce the actual state of the environment, climate modellers have applied 'adjustments' to get the numbers to work out. The result is that climate models are both fundamentally wrong and have been wrongly adjusted.'

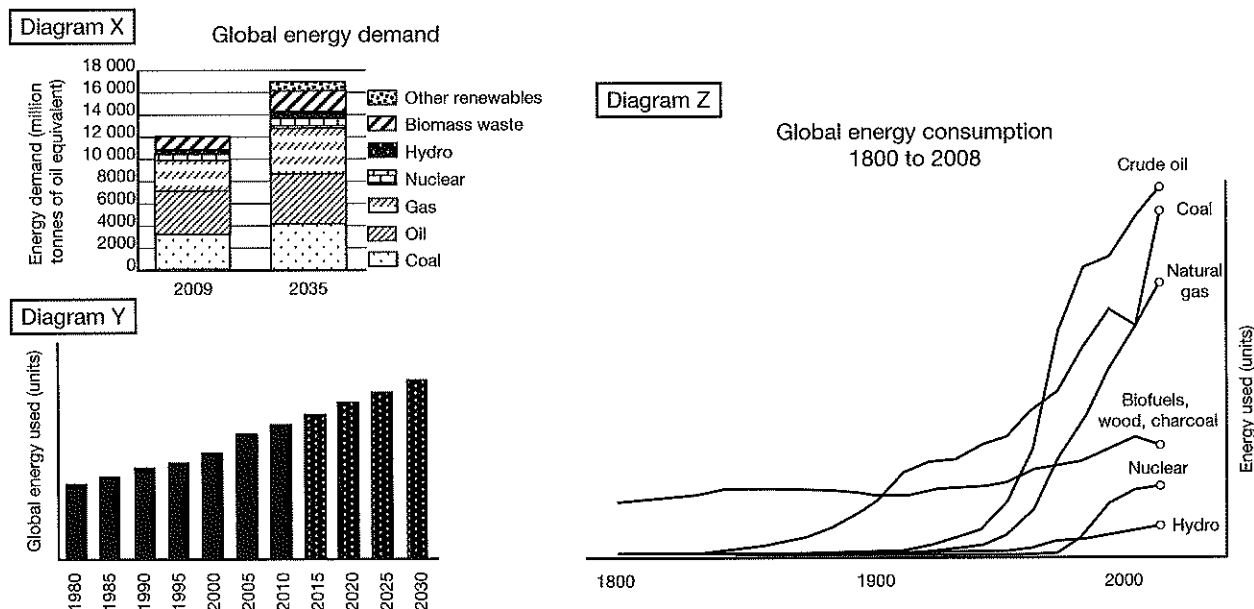
In other words, the models are constructed so as to give these numbers, the numbers are not predictions or calculations of the models (as in the infamous 'we put the physics in and then the answer pops out').

(a) In two short sentences, summarise the two main ideas in this information.

(b) Assess this statement.

1.7.2 Energy sustainability.

1.7.2.1 Consider the three diagrams, X, Y and Z.



- (a) What is the difference between 'Global energy demand' and 'Global energy consumption' and 'Global energy used'?
-
- (b) Why have the three different labels for the y-axis on the diagrams been used?
-
- (c) Which diagram provides us with the best information? Justify your answer.
-
- (d) Suggest a reason for the different shading used on the bars in diagram Y.
-
- (e) Which diagram is the best to use to make predictions into the future? Justify your answer.
-
- (f) What do the diagrams suggest about our ability to have reliable sources of energy far into the future?
-
-
- (g) What is meant by 'energy sustainability'?
-
- (h) Predict the source(s) of global energy if it is to be sustainable into the far distant future. Justify your answer.
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-